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Roll No

EC-7001-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Microwave Engineering

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume suitable data, if required.

1. a) Discuss the general representation of EM field in terms of TEM, TE and TM components.
b) Write and explain the classification of different waveguide structures, also explain its operating principle.
2. a) A rectangular waveguide with dimension of 3×2 cms operates in TM₁₁ mode at 10GHz. Determine the characteristic wave impedance.
b) Discuss some of the properties of Dominant Mode.
3. a) Derive a general band width relation for Micro strip and rectangular waveguides.
b) Discuss the dominant and higher order modes in strip line structure.
4. a) Write and explain the properties of Scattering Matrix of reciprocal network.
b) What do you mean by Directional couplers? Explain their basic derivative parameters.

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5.
 - a) Explain the working principle of two cavity klystron. Discuss some of its applications.
 - b) Draw and explain the block diagram of frequency converters using varactor diodes.
6.
 - a) Discuss some of the properties of ferrites, isolators and phase shifters.
 - b) Derive and explain Manley Rowe power relation.
7.
 - a) Explain the principle of VSWR meter. How it works? Explain with suitable example.
 - b) Discuss about the network analyzer and its use in measurement.
8. Write short notes (any four):
 - a) TWT amplifier
 - b) BARITT
 - c) Gunn Oscillator
 - d) Tee junction of waveguides
 - e) Tuned Detectors
